

Article

The Internal Geometry of Microbial Shoal and Its Reservoir Heterogeneity: Insights from Core Samples of Well X1 in the Pre-Salt Santos Basin

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Abstract

Recently, a substantial quantity of oil and gas has been discovered in the pre-salt Lower Cretaceous microbialite successions of Brazil's Santos Basin, thereby prompting a global surge in research related to microbialites. It has been demonstrated that microbial shoal reservoirs yield the highest hydrocarbon production, with optimal reservoir properties, as evidenced by experience in the field of oilfield production. However, as research progresses, it has become increasingly evident that significant heterogeneity exists in both the lithology and physical properties within microbial shoal bodies. In order to address the identified knowledge gap, the present study employs systematic petrological and petrophysical datasets. These include 30-m continuous core samples, thin-section analyses, routine petrophysical tests and mercury injection capillary pressure (MICP) measurements. The aim is to characterize the internal microfacies architecture and reservoir heterogeneity of microbial shoals. It is imperative to ascertain the principal factors that govern the heterogeneity observed in these reservoirs. This critical step is essential for a comprehensive understanding of the subject matter. The results of the study demonstrate that: the Barra Velha Formation microbial shoals in the Santos Basin can be subdivided into three microfacies, which are delineated from base to top. The foundation of the shoal is the shoal base. The rock composition is dominated by the presence of spherulites, with intracrystalline pores functioning as the primary reservoir spaces. The compositional rocks of the shoal flank are poorly sorted microbial debris, with intergranular and intragranular pores formed by penecontemporaneous dissolution. The sedimentary succession of the shoal core is characterized by well-sorted microbial debris rocks displaying multiple shallowing-upward sequences, with reverse-graded textures. The primary storage space is constituted by fabric-selective pores from penecontemporaneous dissolution, though these are subject to local disruption by destructive silicification. Meanwhile, the microbial shoals demonstrate wide porosity (8.8–26.4%, mean 16.8%) and permeability (0.13–839 mD, mean 169 mD) ranges, thus classifying them as medium-porosity, high-permeability reservoirs. The superimposition of microfacies and diagenetic processes gives rise to considerable reservoir heterogeneity. It is evident that the shoal core microfacies exhibits robust energy and substantial grain size, characteristics that facilitate its exposure above lake level during periods of high-frequency lake-level oscillation. This exposure is further compounded by the influence of atmospheric water dissolution, which remodels the microfacies during the quasi-contemporaneous period. The reservoir quality is optimal, exhibiting the highest proportion of large pores. The reservoir properties of the shoal flank are closely followed by medium and large pores, and those of the shoal base are the worst, with micro and medium pores.



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Keywords: Santos Basin; Lower Cretaceous; microbial shoal; internal geometry; heterogeneity; diagenesis; reservoir quality

1. Introduction

Shoal is defined as a high-energy, wave- or tide-dominated aggregate of sand-sized carbonate particles, primarily forming on platform margins, ramps, or shelves [1,2]. As one of the four major carbonate reservoir types [3,4], statistics from the C&C database reveal that grain shoals constitute approximately 25% of global carbonate reservoirs, with a wide distribution spanning regions such as the Middle East, Central Asia, and the Americas [3,4]. Stratigraphically, these occurrences are distributed across geological periods, except for the Quaternary, and demonstrate concentrated development in the Cretaceous, Jurassic, Carboniferous, and Permian strata. Many studies have been carried out on carbonate shoals by previous researchers, and the following important progress has been made. Based on the study of a modern ooid sand body in the Bahamas, the main rock types and the development and evolution models of ooid sand bodies have been defined [5]. With regard to studies of internal structure, a multitude of classification schemes has been proposed. Based on the sedimentary setting, shoal bodies are divided into two distinct categories, namely intra-platform or marginal-platform shoals [6]. The classification of the shoal is determined by the variations in rock types, sedimentary structures and paleontological types, resulting in the delineation of foreshoal, shoal and backshoal elements [7]. The classification of grain shoals according to particle type encompasses sandy shoals, oolitic shoals, gravel shoals, and other categories [8,9]. The classification of shoals according to the energy and development position of the water body is a subject of ongoing research [10]. The current state of the literature on the subject suggests that shoals can be divided into three categories: high-energy, low-energy, and inter-shoal sea. It is evident that, in accordance with the diagenetic environment, the carbonate shoal is to be divided into two discrete categories: namely, the sparry shoals and the micritic shoals [11]. It has been determined that the shoal within the platform is divided into shoal core, shoal flank and inter-shoal depression elements based on the disparity in particle content and water energy [12]. The academic community holds divergent views on the heterogeneity of shoal deposits. Recent research in the field of sedimentology has indicated that cross-bedded, well-sorted ooid facies are subject to intense cementation, leading to a reduction in porosity and permeability. Conversely, bioturbated, poorly sorted ooid facies have been observed to exhibit higher levels of porosity and permeability [13]. However, alternative theories have been proposed by other scholars. It is argued by these researchers that the relationship between microfacies development within shoal bodies and hydrodynamic energy means that shoal cores tend to form high-quality pinhole reservoirs, while shoal margins predominantly feature thin-bedded grainstone–marl interbeds with less developed dissolution [12].

In recent years, an increasing number of microbial shoals have been reported as oil reservoirs around the globe, including the Oman Intracratonic Basin [14], the Kazakhstan Pre-Caspian Basin [15], the South Asia Upper Indus Basin [16], the Sichuan Basin of China [17], and the Piceance Basin of the United States [18]. Meanwhile, a significant quantity of world-class oil and gas discoveries has been made in Brazil's pre-salt carbonate successions. However, the academic community has yet to reach a consensus on the origin of the BVE Formation carbonate rocks in Brazil's pre-salt layer. The predominant hypotheses encompass microbial origins [19,20], chemical origins [21], and tufa/travertine formations [22,23]. For the purpose of enhancing clarity in subsequent discussions, this study adopts the microbial origin classification scheme [19]. A thorough analysis of hydrocarbon production

data reveals that microbial reef-shoal reservoirs yield the highest output, which has therefore driven increased research focus on microbial reef-shoal systems. Previous research has outlined the pre-salt play and hydrocarbon exploration history in Brazil [24]. It has been proposed that pre-salt carbonate buildups are controlled by two types of carbonate platforms: inter-depression uplift and intra-depression uplift [25]. Furthermore, the stratigraphic stacking patterns and vertical/lateral distribution of microbial reef-shoals have been reproduced through sedimentary forward modelling [26]. Seismic reflection characteristics have been clarified [27,28], facies stacking patterns have been established on cores and depositional models for microbial reef-shoals [29,30], and the correlation between lithology, key diagenetic processes, and physical properties has been analyzed [31]. As exploration and development continue apace, the study of internal heterogeneity in microbial shoal is becoming increasingly important. However, there is a paucity of reports on microfacies composition, superposition style, heterogeneity and its main control factors in microbial shoals [32]. These factors are pivotal in controlling the high production and enrichment of microbial shoal reservoirs. Drilling cores are considered to be the most intuitive and reliable initial data sets reflecting underground geological characteristics [33], with the capacity to directly reflect lithology, structure, and the internal structure of microbial shoals. Given this context, the present study employs the pre-salt Barra Velha Member 100 in the Santos Basin as a case study. The lithological composition and stacking patterns of microbial shoals were analyzed on the basis of 30 m of continuous core sampling and 33 thin sections. Furthermore, by integrating petrophysical analysis of 22 samples and mercury injection capillary pressure (MICP) data from four samples, the heterogeneity of the reservoir in microbial shoals was characterized, and key controlling factors were identified. The research findings may provide references for similar studies.

2. Geological Settings

The Santos Basin, located in the vicinity of southeastern Brazil, was formed during the Early Cretaceous breakup of Gondwana, encompassing an area of approximately 200,000 km². The region under consideration is delimited as follows: to the north by the Cabo Frio Arch (thus separating it from the Campos Basin; to the south by the Florianopolis Platform; to the west by the Precambrian basement uplift of Serra do Mar; and to the east by the thinned continental crust of the Sao Paulo Plateau (Figure 1a) [34]. The basin is structurally divided into five zones, ranging from nearshore to deep-water settings. The following zones have been identified: the Nearshore Depression Zone, the Central Low Uplift Zone, the Offshore Depression Zone, the External High Uplift Zone (where major oilfields and Well X1 are located), and the Abyssal High Uplift Zone (Figure 1a) [35].

The Santos Basin can be subdivided into three distinct stratigraphic units based on vertical stratification: the pre-salt strata, the evaporation strata, and the post-salt strata. The pre-salt sequences are defined as evaporite from the crystalline basement to the middle-late Aptian stage of the Lower Cretaceous, encompassing lacustrine hydrocarbon source rocks and shell limestone of the Balemian stage, and the microbial carbonate reservoir of the Barra Velha Formation (henceforth referred to as the BVE Formation) of the early-middle Aptian stage. It has been determined that microbial carbonate oil and gas fields are currently located in the subhalite series, particularly within the upper 100 members of the BVE, which is recognized as the concentrated distribution horizon of microbial reefs and shoals. The evaporation strata are defined as the substantial evaporite deposits that accumulated during the middle and late Aptian stages, constituting the high-quality regional cap rock within the designated study area. The upper-salt layer is defined as the stratum situated above the Aptian stage. The oil reservoir is predominantly characterized by deep-water turbidite, as illustrated in Figure 1b.

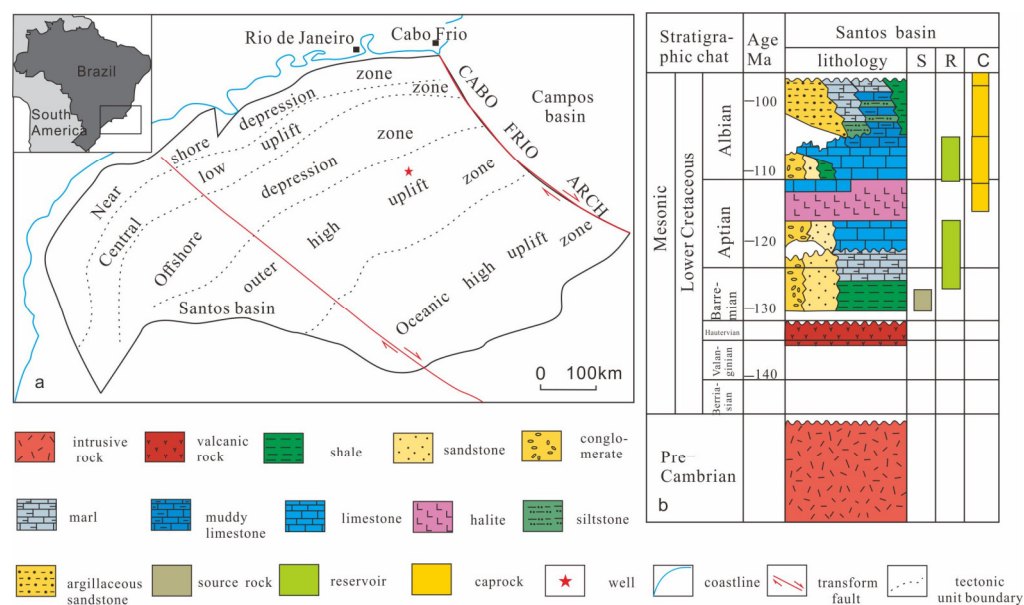


Figure 1. Location and stratigraphic column diagram of the Santos Basin ((a) modified from reference [35–37]; (b) modified from reference [29], in Figure (b), S represents the source rock, R represents the reservoir, and C represents the cap rock).

3. Materials and Methods

3.1. Samples

The pre-salt microbial reef-shoal reservoirs in Brazil are predominantly distributed within the BVE100 interval. The present study focuses on Well X1 and utilizes 30 m of continuous core samples (5063–5093 m) to conduct the following analyses: thin sections were prepared in a total of 75 cases, using blue epoxy resin impregnation. In addition, 22 petrophysical property tests were carried out, along with 4 mercury injection capillary pressure (MICP) experiments. These analyses characterize the lithology, petrophysical properties, internal structure, and heterogeneity of the microbial shoal facies.

3.2. Laboratory Analysis

All sample testing was conducted at the CENPES-Leopoldo A. Miguez de Mello Research and Development Centre of Petrobras, Rio de Janeiro, Brazil. The preparation of cast thin sections involves the initial removal of crude oil, bitumen, and other organic substances from the core samples, followed by the subsequent injection of blue epoxy resin into the pore space under high pressure. The thin section identification was conducted utilizing a Zeiss binocular microscope. Conventional core analysis (including porosity and permeability) was completed based on the helium method under experimental conditions of humidity 45% and temperature 60 °C. The mercury injection capillary pressure (MICP) analysis is a prevalent technique employed in the study of reservoir pore structure. The experiment encompasses sample drying, sample preparation, vacuum degassing, mercury injection and withdrawal, and data correction. All laboratory procedures are in accordance with the American Petroleum Institute.

4. Results

4.1. Internal Geometries of Microbial Shoal

4.1.1. Rock Types of Microbial Shoal

Based on systematic core and thin-section analyses, this study classifies microbial shoal rocks in Brazil's pre-salt sequence into the following types: spherulite, poorly sorted stromatolitic debris grainstones, well-sorted stromatolitic debris grainstones, well-sorted

debris of spherulites, silicified stromatolitic debris rocks, with minor occurrences of well-sorted intraclasts.

Spherulite: It constitutes a distinctive rock type within the pre-salt formations of Brazil. These minerals typically possess diameters ranging from 1 to 2 mm and manifest as sub-spherical forms under microscopic examination. When observed under crossed-polarized light, characteristic cross extinction is exhibited. Intergranular spaces are found to be filled with either authigenic magnesian clay or Mg-silicates, composed of stevensites, saponites, kerolites et al. (Figure 2a) [21,38]. Alternatively, these spaces may be filled with fine- to medium-crystalline euhedral to subhedral dolomite (Figure 2b). As the sequence progresses, there is the observed decrease in the abundance of authigenic magnesian clay/Mg-silicate cements is accompanied by an increase in dolomite cement content.

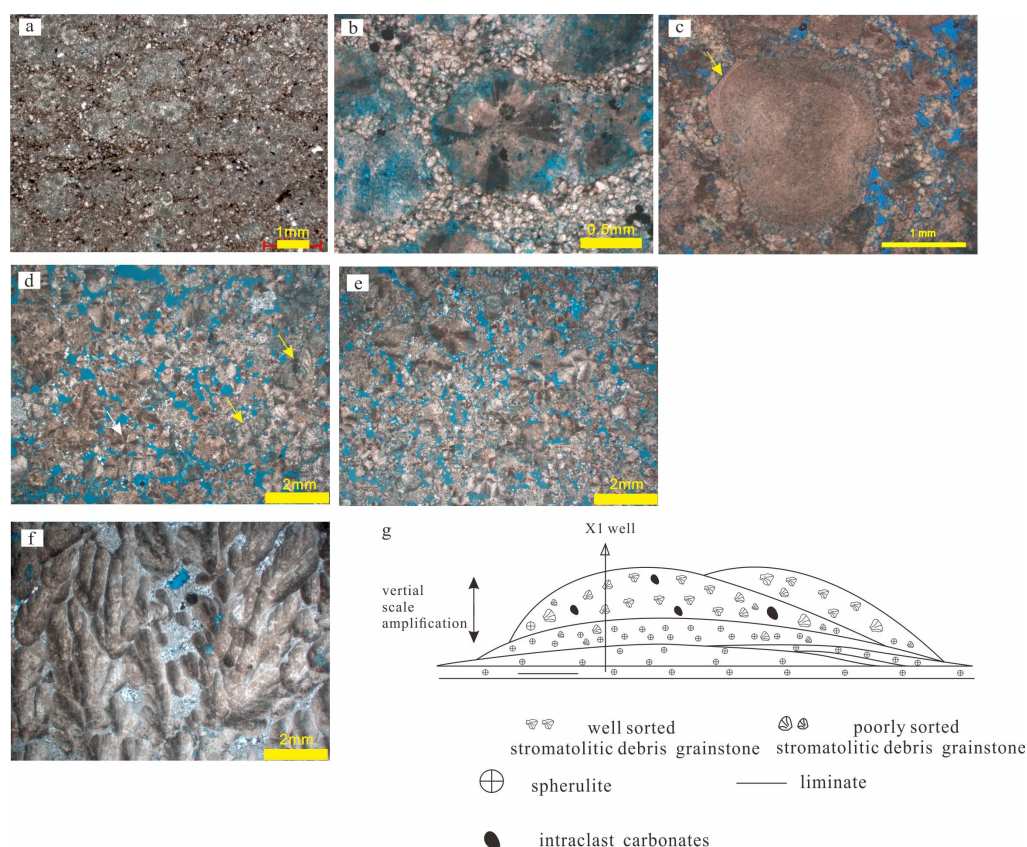


Figure 2. Rock types and internal architecture of the microbial shoal. (a) Spherulite with abundant authigenic magnesian clay or Mg-silicates in intergranular spaces, plane-polarized light (PPL), 5092.8 m; (b) spherulite showing grains with cross-polarized extinction, with intergranular spaces filled by fine- to medium-crystalline euhedral-subhedral dolomite. Cross-polarized light (XPL), 5085 m; (c) poorly sorted stromatolitic debris grainstones primarily composed of stromatolite fragments (yellow arrows: transverse section exhibiting distinct laminations). PPL, 5074.4 m; (d) well-sorted stromatolitic debris grainstones dominated by spherulites (white arrows) and stromatolite fragments (yellow arrows). XPL, 5065.65 m; (e) well-sorted debris of spherulites displaying cross-polarized extinction texture. XPL, 5067.45 m; (f) Silicified stromatolitic debris rocks. XPL, 5065.05 m. (g) conceptual model diagram of the internal geometry of the microbial shoal.

Poorly sorted stromatolitic debris grainstones constitute one of the primary lithologies in microbial shoal environments. It consists of reworked stromatolitic debris. In core samples, it manifests in layered or massive forms (Figure 3). The stromatolitic debris displays variable sizes, with thin sections revealing distinct lamination structures in cross-sectional views (Figure 2c).

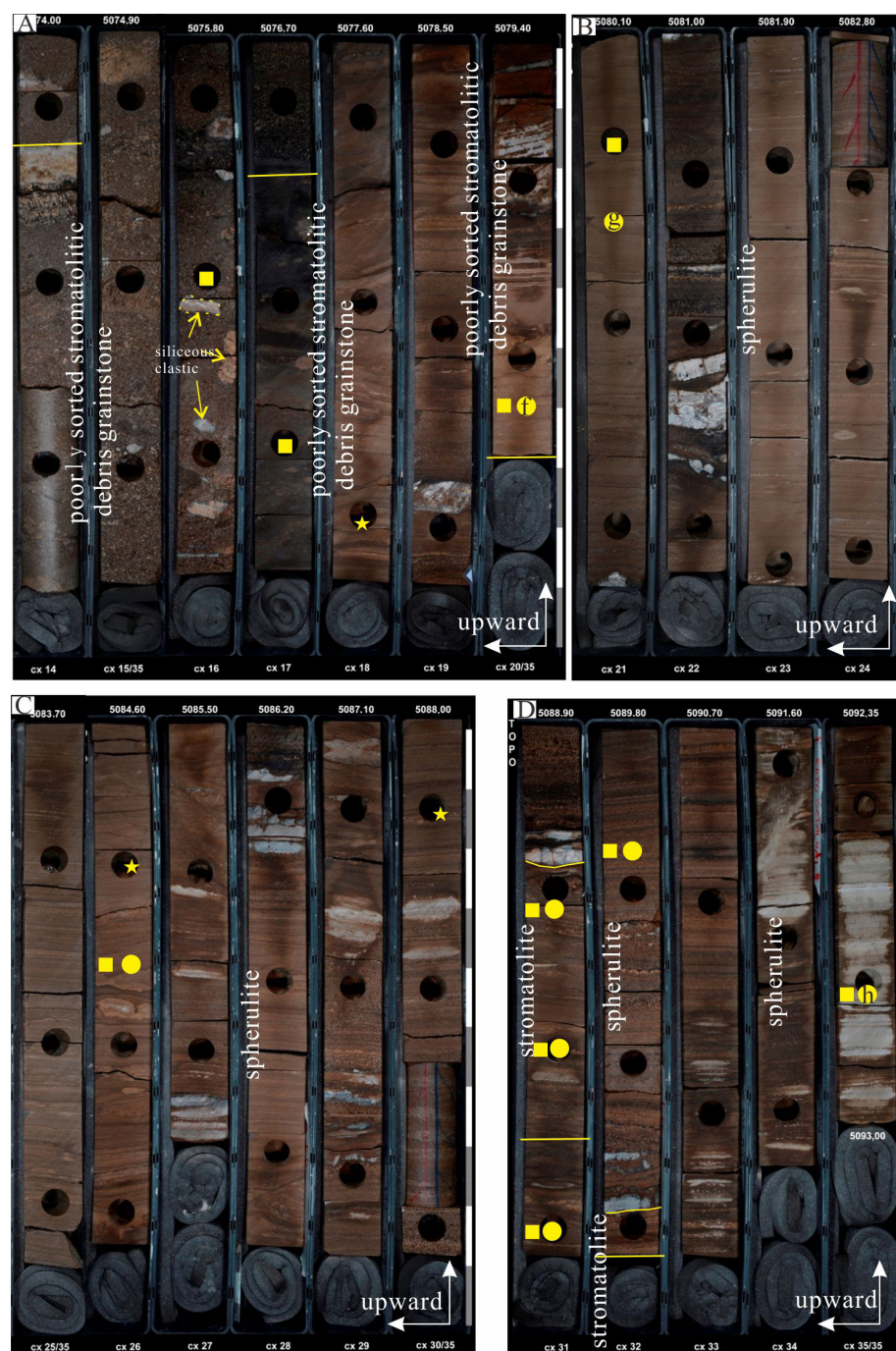


Figure 3. Lithological assemblage of the microbial shoal flank (A) and shoal base (B–D). The locations of thin sections are indicated by yellow circles, those of petrophysical sampling points by yellow squares, and those of mercury injection capillary pressure (MICP) sampling points by yellow five-pointed stars. Additionally, it should be noted that each segment of the color bar on the right of the figure represents 10 cm. The thin sections for f, g, and h are provided in Figure 4.

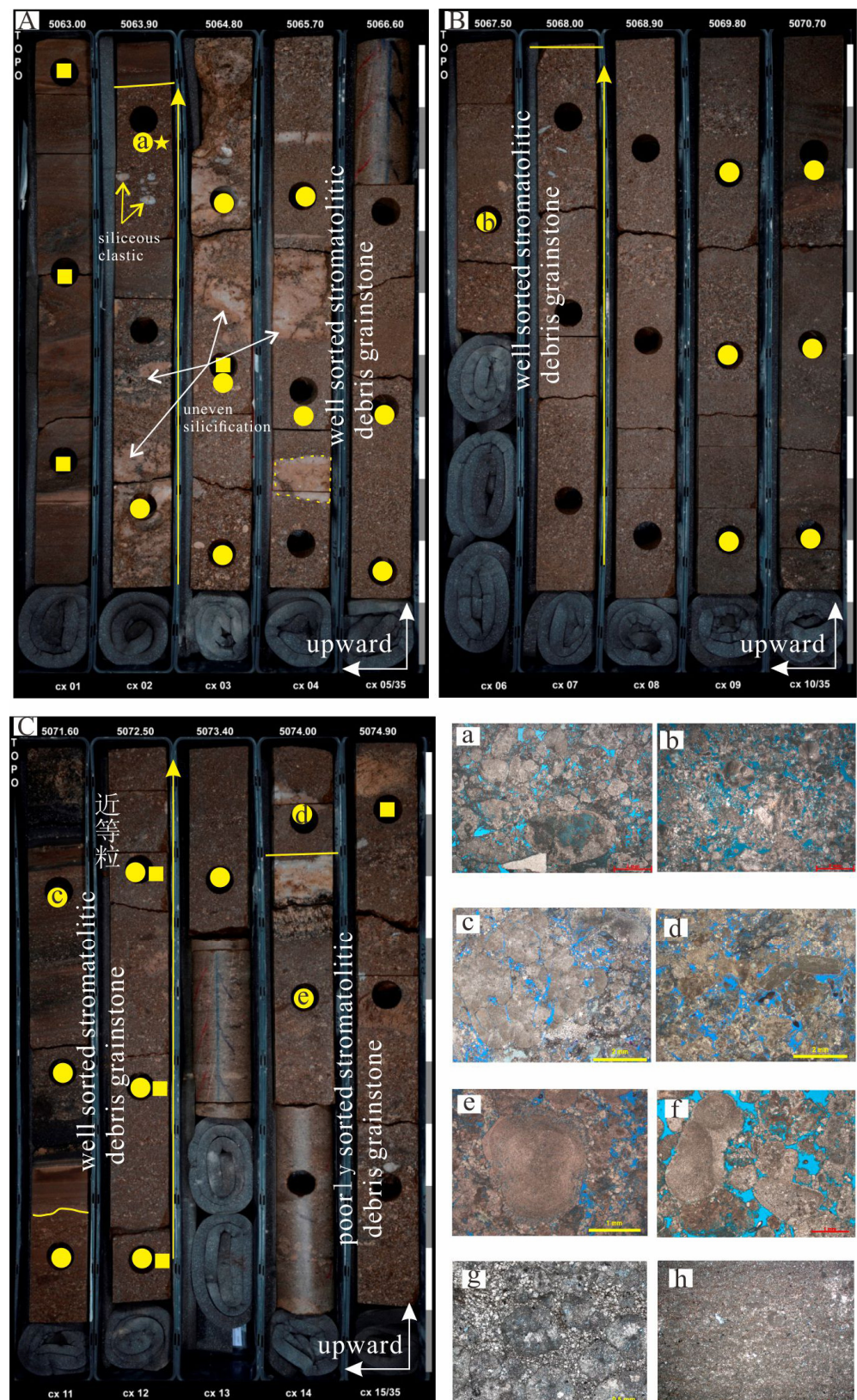


Figure 4. Lithological assemblage of the microbial shoal core (A–C,a,b). The locations of thin sections are indicated by yellow circles, those of petrophysical sampling points by yellow squares, and those of mercury injection capillary pressure (MICP) sampling points by yellow five-pointed stars. Additionally, it should be noted that each segment of the color bar on the right of the figure represents 10 cm. The positions of thin sections (a–e) correspond to locations (a–c) in the core.

Well-sorted stromatolitic debris grainstones: As another primary lithology in microbial shoals, these rocks are formed by fragmented stromatolites. In core samples, the prevalence of massive forms is predominant (Figure 3). The stromatolitic debris displays approximately uniform sizes, with partially exposed surfaces revealing concentric growth band structures under microscopic examination (Figure 2d).

Well-sorted debris of spherulites: These are formed by fragmented spherulites. In core samples, they commonly occur interbedded with well-sorted stromatolitic debris rocks. Microscopic examination reveals the presence of spherulite, as illustrated in Figure 2e, which exhibits the characteristic cross-extinction structure.

Silicified stromatolitic debris rocks: In core samples, they manifest as light gray and occur as siliceous lithoclasts, unevenly silicified patches, siliceous bands, and silicified layers (Figure 3). Microscopic examination reveals silicification exhibiting characteristic mimetic replacement patterns, with both stromatolitic structures and cements remaining well-preserved (Figure 2f).

4.1.2. Lithological Assemblages and Stacking Patterns of Microbial Shoal

Based on the analysis of core and thin sections, microbial shoals are classified into three lithological assemblages according to vertical stacking patterns: shoal base, shoal core, and shoal flank. Their configuration model is illustrated in Figure 2g. Shoal base: The site is located at the lowest point of the shoal, with an approximate thickness of 13 m (5080.1–5093.0 m; Figure 3). This interval was formed in a deep-water, low-energy environment characterized by the presence of spherulite. The basal section displays evidence of cementation by authigenic magnesian clay or Mg-silicates, with an upward progression in the dolomite cement content, resulting in the formation of a sheetlike basement. Stromatolitic debris is present locally.

Shoal flank: Located along the periphery of the shoal body in relatively deep-water settings with a thickness of approximately 6 m (5074.15–5080.10 m; Figure 3). This low-energy interval comprises mixed stromatolites, spherulites and intraclasts, which are characterized by poorly sorted textures. A vertical transition in facies is evident, characterized by a transition from laminated, poorly sorted microbial grainstone to massive, poorly sorted microbial grainstone. In the substantial unit, centimeter-scale chert clasts demonstrate orientations that range from sub-horizontal to near-vertical. The succession exhibits an overall fining-upward sequence with positive grading.

Shoal core: Positioned in the central part of the shoal body, with an approximate thickness of 11 m (5063.0–5074.15 m; Figure 4). The sedimentary rock composition is characterized by the presence of moderately sorted clasts, which include stromatolites, spherulitic microbialites, and intraclasts. In the vertical direction, the structure comprises multiple coarsening-upward sequences, as illustrated by the yellow arrows in Figure 3. The upper portion of each cycle displays silicification features, including: The presence of silica clasts, selective silicification, and silica crusts has been identified (Figure 4). Thin-section analysis reveals complete silicification of stromatolites and cements (Figure 4).

4.2. Heterogeneity of Microbial Shoal Reservoir

4.2.1. Types of Microbial Shoal Reservoir Spaces

The spaces within microbial shoals which function as reservoirs are characterized by a preponderance of fabric-selective porosity types. These include intergranular pores, intergranular dissolution pores, intragranular dissolution pores, and moldic pores. Meanwhile, it has been observed that there are minor occurrences of intracrystalline pores.

Intergranular pores, intergranular dissolution pores, intragranular dissolution pores, and moldic pores constitute the most significant reservoir spaces in microbial shoals. A mi-

microscopic examination reveals the following: intergranular pores (Figure 5a), intergranular dissolution pores formed by the enlargement of intergranular spaces (Figure 5b), intragranular dissolution pores (Figure 5c), coexisting intergranular and intragranular dissolution pores (Figure 5d), and sporadic moldic pores in partially silicified stromatolite intervals (Figure 5e). These reservoir spaces exhibit two defining characteristics: fabric-selectivity and facies-dependence, predominantly developed in grain-rich carbonate microfacies of the shoal core and shoal flank.

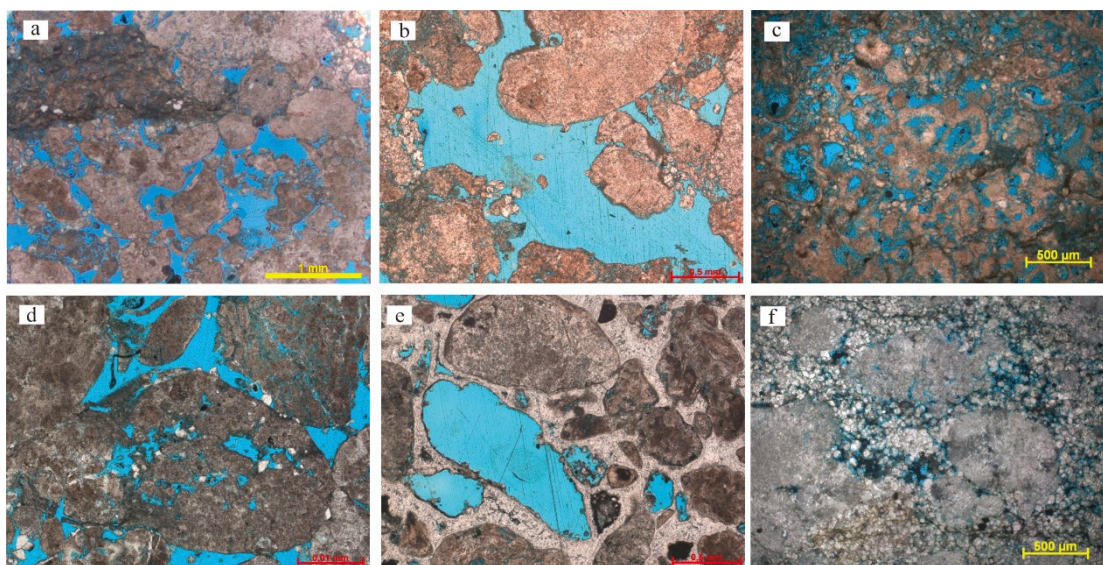


Figure 5. Reservoir space types in the microbial shoal. (a) Interparticle pores and interparticle dissolution pores; blue-epoxy-impregnated thin section, plane-polarized light, 5073.6 m. (b) interparticle dissolution pores, blue-epoxy-impregnated thin section, plane-polarized light, 5065.65 m. (c) intraparticle dissolution pores, blue-epoxy-impregnated thin section, plane-polarized light, 5063.7 m. (d) interparticle pores and intraparticle dissolution pores, blue-epoxy-impregnated thin section, plane-polarized light, 5070 m. (e) moldic pores, blue-epoxy-impregnated thin section, plane-polarized light, 5064.65 m. (f) intercrystalline pores, blue-epoxy-impregnated thin section, cross-polarized light, 5064.65 m.

Intracrystalline pores are predominantly developed in the relatively deep-water shoal base microfacies, which is characterized by a high abundance of authigenic Mg-rich clays or Mg-silicates. Petrographic observations indicate that the cements between spheroidal microbialite grains are composed of fine- to medium-crystalline euhedral to subhedral dolomite, with well-developed intracrystalline pores (Figure 5f). The presence of yellowish-brown to black discoloration in the pores is indicative of subsequent hydrocarbon invasion.

4.2.2. Properties of Microbial Shoal

A series of systematic petrophysical tests was conducted on 21 samples, revealing that microbial shoal reservoirs exhibit favorable property correlations but strong overall homogeneity (Figure 6). The porosity of the samples ranges from 8.8% to 26.4%, with an average of 16.8%, indicating a unimodal distribution (Figure 7a). To elaborate further: as demonstrated in Figure 7, 10% of the samples exhibited porosity within the range of 4% to 12%. A substantial proportion, constituting 62%, exhibited porosity levels ranging from 12% to 20% (Figure 7c). Concurrently, the range of permeability distribution was found to span from 0.13 mD to 839 mD, with an average value of 169.9 mD, indicative of a multimodal distribution (Figure 7b). As demonstrated in Figure 7, the permeability of samples ranging from 0.1 to 1 mD accounts for 14%, those ranging from 1 to 10 mD account for 29%, those ranging from 10 to 100 mD account for 19%, and those over 100 mD

account for 38% (Figure 7d). The overall permeability of the reservoir can be classified as medium-porosity and high-permeability [39].

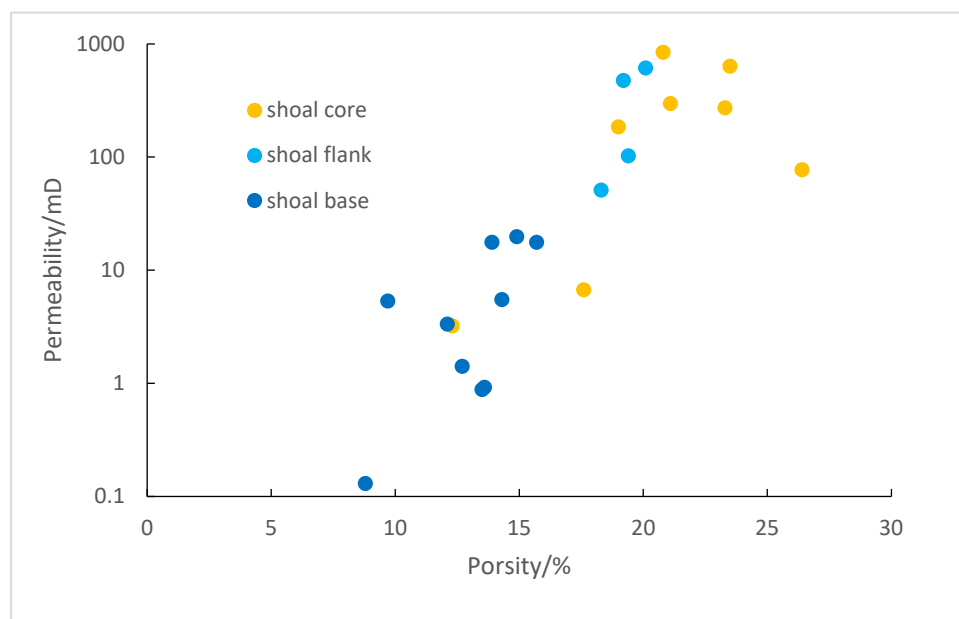


Figure 6. Intersection diagram of microfacies porosity and permeability of microbial shoal.

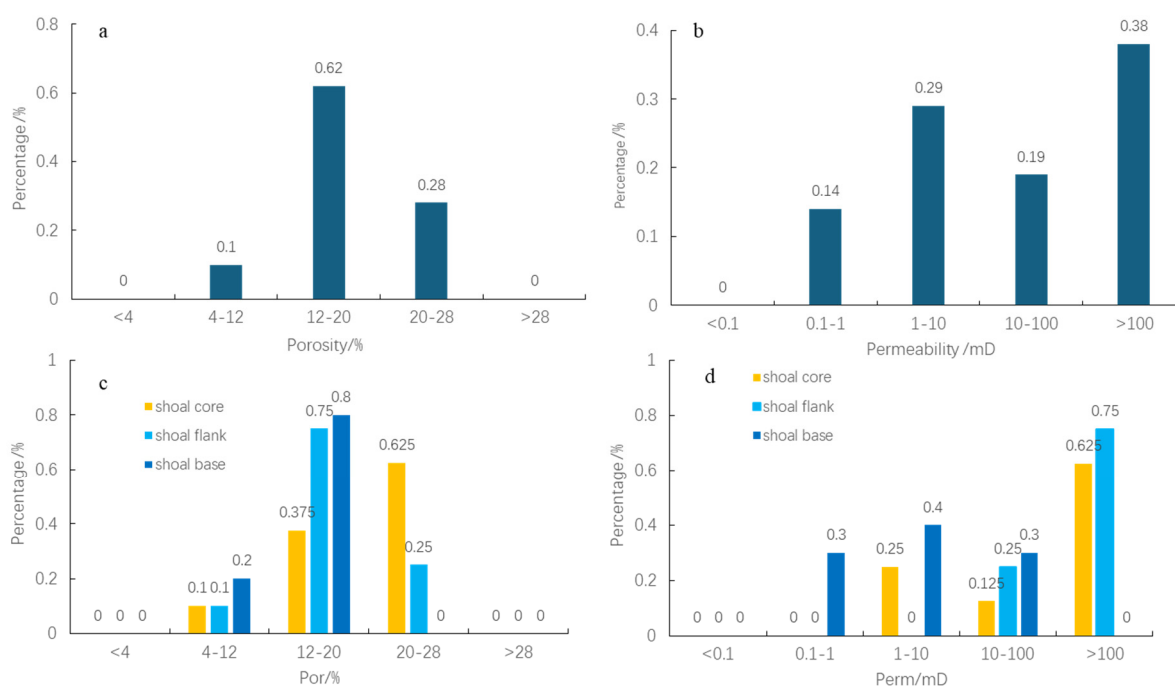


Figure 7. Porosity and permeability distribution interval of microbial shoal. (a) Porosity distribution ranges of microbial shoals, (b) Permeability distribution ranges of microbial shoals, (c) Porosity distribution ranges of different microfacies within microbial shoals, (d) Permeability distribution ranges of different microfacies within microbial shoals.

5. Discussion

Within the academic community, there is a broad consensus that carbonate reef-shoal reservoirs are archetypal facies-controlled reservoir types [40,41]. Furthermore, due to the diagenetic sensitivity of carbonate minerals such as calcite and dolomite [42,43], diagenesis plays a critical role in reservoir heterogeneity studies. The subsequent section will examine

the dominant controlling factors of heterogeneity in microbial shoal reservoirs from the perspectives of microfacies and diagenesis.

5.1. The Influence of Microfacies on the Heterogeneity of Microbial Shoal Reservoirs

In order to analyze the differences in pore structure among the various microfacies, samples were collected from the microbial shoal core, shoal flank, and shoal base, respectively, for the purpose of conducting a mercury injection experiment analysis. The findings demonstrated that a gradual coarsening process occurred in the grain size of the shoal base, shoal flank and shoal core as the water depth decreased and the lime content reduced. Concurrently, the sorting quality exhibited an enhancement, the pore throats underwent a coarsening process, and the corresponding threshold pressure underwent a gradual decrease (Figure 8). Moreover, quantitative analysis of the differences in pore structure of the various microfacies of the microbial shoal demonstrates that macropores are the predominant microfacies in the shoal core, accounting for 68.19% of the total, followed by mesopores with 27.19%, and micropores with a minimal contribution of only 4.62% (Figure 9). The proportion of macropores, mesopores and micropores in the upper part of the shoal base is only 1.75%, 58.97% and 39.28% respectively. Despite the presence of a significant proportion of large pores in the lower part of the shoal base, measuring at 12.86%, there is a substantial decline in the proportion of mesopores to 38.56%, accompanied by a notable increase in the proportion of micropores to 48.58%. This observation indicates a heightened level of heterogeneity within the shoal base (Figure 10).

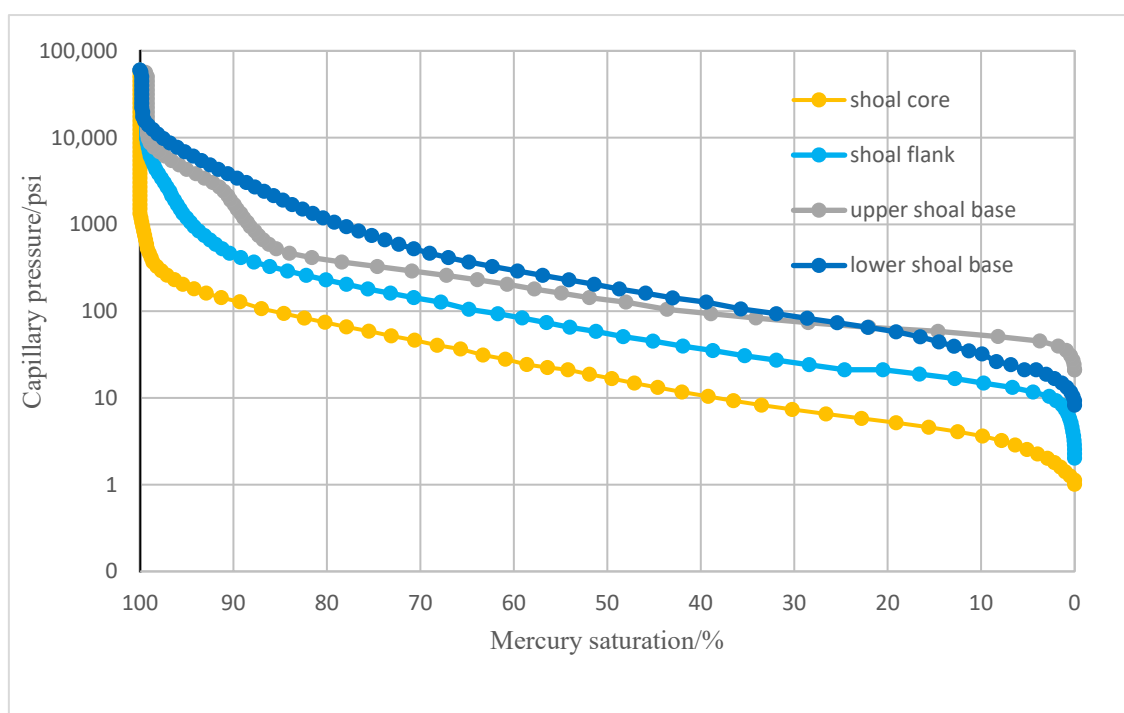


Figure 8. Capillary pressure curves of different microfacies in microbial shoal (the sampling locations are indicated by yellow pentagrams in Figure 3).

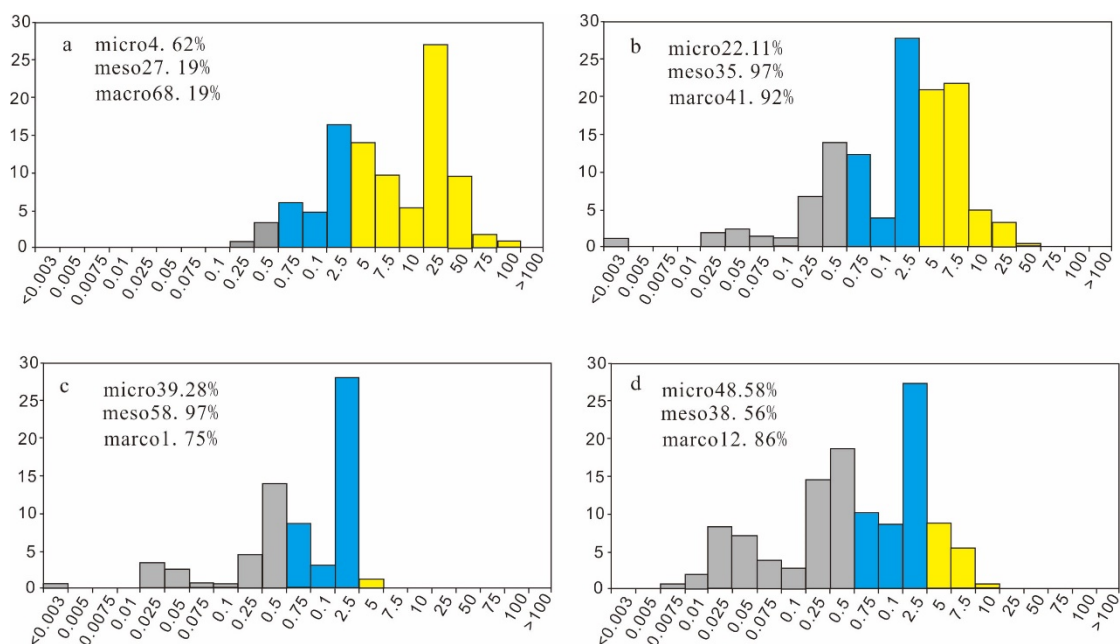


Figure 9. Dominant control of microbial shoal microfacies on pore architecture: (a) shoal core, (b) shoal flank, (c) upper shoal base, (d) lower shoal base. Note that in this figure, gray color bar represents micropores, blue represents mesopores, and yellow represents macropores.

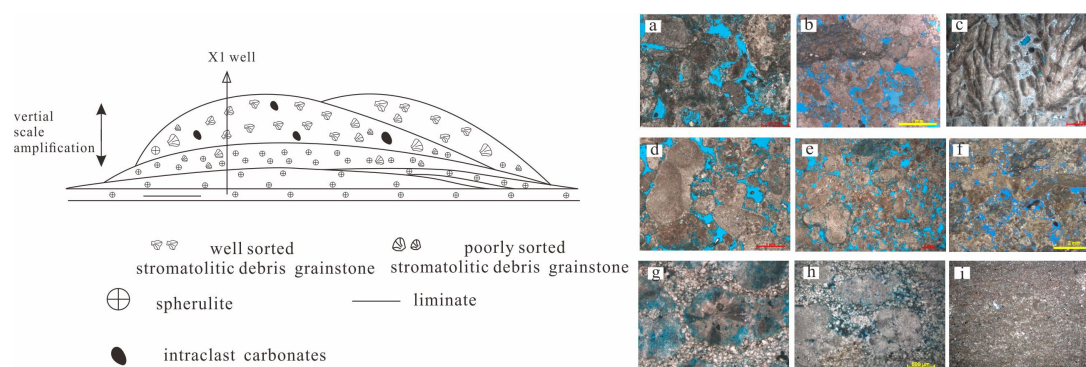


Figure 10. Diagenetic difference in microfacies in microbial shoal. (a) Shoal Core: interparticle pores and interparticle dissolution pores formed by penecontemporaneous dissolution, blue-epoxy-impregnated thin section, plane-polarized light, 5070 m; (b) shoal Core: interparticle dissolution pores formed by penecontemporaneous dissolution, cast thin section with the pores filled with blue-dyed resin. PPL, 5087.5 m; (c) shoal Core: silicified stromatolitic grainstone fragment formed by penecontemporaneous silicification and dissolution, where stromatolite fabric and cement textures are well-preserved, exhibiting mimetic replacement features, casting thin section with the pores filled with blue-dyed resin, PPL, 5064.65 m; (d) shoal flank: unequal-sized stromatolitic grainstone with interparticle pores formed by penecontemporaneous dissolution, casting thin section, with the pores filled with blue-dyed resin, PPL, 5080 m; (e) shoal flank: unequal-sized stromatolitic grainstone, with interparticle pores and interparticle dissolution pores formed by penecontemporaneous dissolution, casting thin section with the pores filled with blue-dyed resin, PPL, 5080 m; (f) shoal flank: unequal-sized stromatolitic grainstone, with interparticle pores and interparticle dissolution pores formed by penecontemporaneous dissolution, casting thin section with the pores filled with blue-dyed resin, PPL, 5074.1 m; (g) shoal base (upper part): spheroidal microbialite, with interparticle euhedral to subhedral dolomite cement, casting thin section with the pores filled with blue-dyed resin, cross-polarized light, 5085 m; (h) shoal base: spheroidal microbialite, casting thin section with the pores filled with blue-dyed resin, cross-polarized light, 5090 m; (i) shoal base: micritic limestone, cast thin section with the pores filled with blue-dyed resin, PPL, 5092.8 m.

5.2. The Influence of Diagenesis on the Heterogeneity of Microbial Shoal Reservoirs

It is evident that diagenesis exerts a pivotal function in the development of carbonate reservoirs. A comprehensive analysis has been undertaken, the results of which indicate considerable differences in diagenetic processes across the diverse microfacies of the microbial shoal. The differences can be outlined as follows:

The shoal core is distinguished by its strong hydrodynamic energy and shallow water depth, resulting in the formation of accumulations of massive, sub-equigranular stromatolitic grainstone and spherulite microbialites. The most typical diagenetic processes observed in the shore core are intensive penecontemporaneous dissolution and localized early-stage silicification. It is evident that the shallow water setting and concurrent conditions resulted in the intermittent exposure of the shoal core above lake level. This exposure rendered the core susceptible to meteoric water leaching. This process resulted in the formation of fabric-selective pores, which include interparticle pores, interparticle dissolution pores, intraparticle pores, and moldic pores [44] (Figure 10a,b). Concurrently, a range of silicification events is evident within the shoal core. It is deduced from the mimetic replacement textures and the intergrowth of siliceous clasts with stromatolitic grainstone that the silicification event occurred early and was probably formed by penecontemporaneous metasomatism of silica-rich fluids. This phase of silicification is indicative of a destructive diagenetic process, which infilled early-stage pores, such as interparticle pores, while performing mimetic replacement (see Figure 10c).

The shoal flank is located on the periphery of the shoal body, with a water depth that is intermediate between that of the shoal core and the shoal base. The site's composition is principally constituted of stromatolitic fragments of unequal size and spherical microbialites, characterized by weak penecontemporaneous dissolution. The primary reservoir space is constituted by fabric-selective pores, encompassing interparticle pores, interparticle dissolution pores, and intraparticle dissolution pores (Figure 10d–f). It is hypothesized that these pores were formed by penecontemporaneous dissolution during meteoric water leaching.

The shoal base is distinguished by its increased water depth, with predominant lithologies comprising laminites, micritic limestone and spheroidal microbialites (Figure 10g–i). Distinct from the shoal core and shoal flank, typical dolomitization is observed in the shoal base. So, the interparticle spaces are filled with fine- to medium-crystalline euhedral to subhedral dolomite cement, which is often associated with authigenic Mg-rich clays or Mg-silicates. Despite the absence of consensus within academia regarding the presence of microbial activity during the formation of these authigenic Mg-rich clays or Mg-silicates [45–47], the associated dolomite and reservoir space have been the subject of study. The hypothesis that dolomitization occurred prior to significant compaction is predicated on the premise that dolomite mimics the wavy structure that is characteristic of the authigenic Mg-rich clays or Mg-silicates. Furthermore, it is hypothesized that the formation of the reservoir space is contingent on the dissolution of the authigenic Mg-rich clays or Mg-silicates [46,48].

6. Conclusions

Based on systematic petrographic and petrophysical analyses, the internal geometry of microbial shoals and the main controlling factors of their heterogeneity have been identified. The primary conclusions that can be drawn from this analysis are as follows:

- (1) The primary lithologies of the microbial shoal include: spherulite, poorly sorted stromatolitic debris grainstones, well-sorted stromatolitic debris grainstones, well-sorted debris of spherulites, and silicified stromatolitic debris rocks, with minor occurrences of well-sorted intraclasts.

- (2) Based on lithological assemblages and stacking relationships, the internal architecture of the microbial shoal is subdivided into three distinct components: the shoal core, shoal flank and the shoal base. The shoal core, located at the highest point of the shoal body, is primarily composed of well-sorted stromatolitic debris grainstones. The shoal flank, situated along the periphery of the shoal body, is primarily composed of poorly sorted stromatolitic debris grainstones. The shoal base, located at the nadir of the shoal body's topography, is characterized by the predominance of spherulites. Ultimately, a stacking model of the shoal base, shoal core, and shoal flank was established.
- (3) It was determined that microfacies and diagenesis were the primary controls on microbial shoal heterogeneity. The shoal core displays a high level of hydrodynamic energy, as evidenced by the presence of coarse grain size and optimal sorting. In this instance, penecontemporaneous dissolution is most evident, leading to reservoirs that are predominantly characterized by macroporosity and optimal petrophysical properties.

The shoal flank is distinguished by a reduction in hydrodynamic energy and an enhancement in sorting. Penecontemporaneous dissolution has been found to be moderately developed, yielding reservoirs that exhibit predominantly meso- to macroporosity and intermediate petrophysical quality.

The shoal base is located at the lowest topographic point within the shoal body. In this area, penecontemporaneous dissolution is poorly developed. It is evident that dissolution pores, formed by the dissolution of authigenic Mg-rich clays or Mg-silicates, constitute the primary reservoir space. This, in turn, results in the poorest possible reservoir quality.

- (4) Given that the dataset for this study is limited to a single well, the proposed internal geometry of microbial shoals (i.e., shoal base, shoal flank, and shoal core) represents an idealized, near-complete geological model. In future practical research, due to variations in drilling locations and the presence of unconformities, microbial shoals may develop distinct internal assemblages. A comprehensive analysis integrating geological, well-log, and seismic data will be required to decipher these internal configuration patterns.

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